

NUMBER THEORY PROJECT NOTES

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1. PRELIMINARY STATEMENTS

Define

$$r_2(n) = \#\{a, b \in \mathbb{Z}_{\geq 0} : n = a^2 + b^2\}$$

be the number of ways to write $n \geq 1$ as a sum of two positive squares and

$$r_4(n) = \#\{a, b, c, d \in \mathbb{Z}_{\geq 0} : n = a^2 + b^2 + c^2 + d^2\}$$

be the number of ways to write $n \geq 1$ as a sum of four positive squares. (Note that if we want $a, b, c, d \in \mathbb{Z}$, then because of the signs $(-a)^2 = a^2$ we get our $r_2(n)$ multiplied by 8 and $r_4(n)$ multiplied by 4).

Throughout the paper we assume that if f is a multiplicative function and $x \notin \mathbb{N}$, then $f(x) = 0$, i.e. $\sigma(n/4) = 0$ for n not divisible by 4.

The classical Jacobi formulas state that

$$r_2(n) = \sum_{d|n} \chi_4(d), \quad r_4(n) = \sum_{\substack{d|n \\ 4 \nmid d}} d = \sigma(n) - \sigma(n/4),$$

where $\sigma(n)$ is the sum-of-divisors function given by

$$\sigma(n) = \sum_{d|n} d$$

and χ_4 is the non-principal Dirichlet character modulo 4, that is χ_4 is a completely multiplicative function with values on primes

$$\chi_4(p) = \begin{cases} 0, & \text{if } p = 2 \\ 1, & \text{if } p \equiv 1 \pmod{4}, \\ -1, & \text{if } p \equiv -1 \pmod{4}. \end{cases}$$

Theorem (Gauss circle problem). *As $x \rightarrow \infty$ we have*

$$(1) \quad \sum_{n \leq x} r_2(n) = \frac{\pi}{4}x + O(\sqrt{x}).$$

Theorem (Lagrange four squares theorem on average). *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} r_4(n) = \frac{\pi^2}{8}x^2 + O(x).$$

Theorem (Mertens' theorems). *As $x \rightarrow \infty$ we have*

$$(2) \quad \sum_{p \leq x} \frac{\log p}{p} = \log x + O(1),$$

$$(3) \quad \sum_{p \leq x} \frac{1}{p} = \log \log x + A + O\left(\frac{1}{\log x}\right),$$

$$(4) \quad \sum_{\substack{p \leq x \\ p \equiv 1 \pmod{4}}} \frac{1}{p} = \frac{1}{2} \log \log x + A + O\left(\frac{1}{\log x}\right),$$

where $A = 0.26 \dots$ is Meissel–Mertens constant.

As a corollary we obtain the following statement that will be used later.

Lemma 1. *As $x \rightarrow \infty$ we have*

$$\sum_{pq \leq x} \frac{1}{pq} = (\log \log x)^2 + A \log \log x + O(1),$$

where $A = 0.26 \dots$ is Meissel–Mertens constant.

Proof. We use (3) and write

$$\begin{aligned} \sum_{pq \leq x} \frac{1}{pq} &= \sum_{p \leq x} \frac{1}{p} \sum_{q \leq x/p} \frac{1}{q} = \sum_{p \leq x} \frac{1}{p} \left(\log \log \frac{x}{p} + A + O\left(\frac{1}{\log(x/p)}\right) \right) \\ &= \log \log x \sum_{p \leq x} \frac{1}{p} + \sum_{p \leq x} \frac{1}{p} \log \left(1 - \frac{\log p}{\log x} \right) + O\left(\sum_{p \leq x} \frac{1}{p \log(x/p)} \right) \\ &= (\log \log x)^2 + A \log \log x + O(1) + S_3(x) + S_4(x), \end{aligned}$$

where

$$S_3(x) = \sum_{p \leq x} \frac{1}{p} \log \left(1 - \frac{\log p}{\log x} \right), \quad S_4(x) \ll \frac{1}{\log x} \sum_{p \leq x} \frac{1}{p} \left(1 - \frac{\log p}{\log x} \right)^{-1}.$$

Therefore, it suffices to bound $S_3(x)$ and $S_4(x)$. We begin with $S_3(x)$ and write

$$S_3(x) = \sum_{p \leq x} \frac{1}{p} \log \left(1 - \frac{\log p}{\log x} \right) \ll \sum_{p \leq x} \frac{\log p}{p} \frac{1}{\log x} \ll 1,$$

where we used Mertens' theorem (2) and the fact that

$$\log(1 - y) \ll y \text{ for } y = \frac{\log p}{\log x} \in (0, 1).$$

So $S_3(x) = O(1)$. Now we similarly evaluate $S_4(x)$.

$$S_4(x) = \frac{1}{\log x} \sum_{p \leq x} \frac{1}{p} \left(1 - \frac{\log p}{\log x} \right)^{-1} \ll \frac{1}{\log x} \sum_{p \leq x} \frac{1}{p} \ll \frac{\log \log x}{\log x} \log \log x,$$

where we used the fact that

$$\frac{1}{1 - y} = \sum_{k=0}^{\infty} y^k \ll 1$$

when $|y| < 1$ which is true in our case, setting $y = \log p / \log x$. Thus, $S_4(x) = O(1)$ and we are done. $\square$

Lemma 2. *As $x \rightarrow \infty$ we have*

$$\sum_{d \leq x} \frac{1}{d} = \log x + O(1).$$

Proof. By partial summation we have

$$\begin{aligned} \sum_{d \leq x} \frac{1}{d} &= \frac{1}{x} \sum_{d \leq x} 1 + \int_1^x \left(\sum_{d \leq t} 1 \right) \frac{1}{t^2} dt = \frac{1}{x} (x + O(1)) + \int_1^x (t + O(1)) \frac{dt}{t^2} \\ &= 1 + O(1/x) + \int_1^x \frac{dt}{t} + O \left(\int_1^x \frac{dt}{t^2} \right) = \log x + O(1). \end{aligned}$$

$\square$

Define

$$\omega(n) = \sum_{p|n} 1, \quad \omega_*(n) = \sum_{\substack{p|n \\ p \equiv 1 \pmod{4}}} 1.$$

Lemma 1. *As $x \rightarrow \infty$ we have*

$$\begin{aligned}\sum_{n \leq x} \omega(n) &= x \log \log x + Ax + O\left(\frac{x}{\log x}\right), \\ \sum_{n \leq x} \omega_*(n) &= \frac{1}{2}x \log \log x + Ax + O\left(\frac{x}{\log x}\right),\end{aligned}$$

where $A = 0.26 \dots$ is Meissel–Mertens constant.

Proof. For both claims we use the definitions and Mertens' theorems

$$\begin{aligned}\sum_{n \leq x} \omega(n) &= \sum_{n \leq x} \sum_{p|n} 1 = \sum_{p \leq x} \sum_{\substack{n \leq x \\ n \equiv 0 \pmod{p}}} 1 = \sum_{p \leq x} \left(\frac{x}{p} + O(1)\right) \\ &= x \sum_{p \leq x} \frac{1}{p} + O\left(\frac{x}{\log x}\right) = x \log \log x + Ax + O\left(\frac{x}{\log x}\right),\end{aligned}$$

where in the last line we used (3).

Similarly, for ω_* we have

$$\begin{aligned}\sum_{n \leq x} \omega_*(n) &= \sum_{n \leq x} \sum_{p|n, p \equiv 1 \pmod{4}} 1 = \sum_{\substack{p \leq x \\ p \equiv 1 \pmod{4}}} \sum_{\substack{n \leq x \\ n \equiv 0 \pmod{p}}} 1 \\ &= \sum_{\substack{p \leq x \\ p \equiv 1 \pmod{4}}} \left(\frac{x}{p} + O(1)\right) = x \sum_{\substack{p \leq x \\ p \equiv 1 \pmod{4}}} \frac{1}{p} + O\left(\frac{x}{\log x}\right) \\ &= \frac{1}{2}x \log \log x + Ax + O\left(\frac{x}{\log x}\right),\end{aligned}$$

where in the last line we used (4). □

Lemma 3. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} \frac{\mu^2(n)}{n} = \frac{6}{\pi^2} \log x + O(1)$$

Proof. We begin by computing $M(x) = \sum \mu^2(n)$.

$$\begin{aligned}M(x) &= \sum_{n \leq x} \mu^2(n) = \sum_{n \leq x} \sum_{d^2|n} \mu(d) = \sum_{d \leq \sqrt{x}} \mu(d) \sum_{\substack{n \leq x \\ d^2|n}} 1 \\ &= \sum_{d \leq \sqrt{x}} \mu(d) \left(\frac{x}{d^2} + O(1)\right) \\ &= x \left(\sum_{d \leq \sqrt{x}} \frac{\mu(d)}{d^2}\right) + O\left(\sum_{d \leq \sqrt{x}} \mu(d)\right) \\ &= x \sum_{d \leq \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x})\end{aligned}$$

then via Dirichlet series we have that this is

$$x \frac{6}{\pi^2} + O(\sqrt{x})$$

Finally we can apply Abel Summation with $A(x) = \sum \mu^2(n)$ and $f(x) = 1/x$ to obtain the desired result. Upon denoting $S(x) = \sum_{n \leq x} \mu^2(n)/n$ we get

$$\begin{aligned} S(x) &= \left(x \frac{6}{\pi^2} + O(\sqrt{x}) \right) \frac{1}{x} + \int_1^x t \left(\frac{6}{\pi^2} + O(\sqrt{t}) \right) \frac{1}{t^2} dt \\ &= \frac{6}{\pi^2} + O(1/\sqrt{x}) + \int_1^x \frac{6}{t\pi^2} + O(1/t^{3/2}) dt \\ &= \frac{6}{\pi^2} \log x + \frac{6}{\pi^2} + O\left(\frac{1}{\sqrt{x}}\right) = \frac{6}{\pi^2} \log x + O(1) \end{aligned}$$

□

Lemma 4. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} \frac{2^{\omega(n)}}{n} = \frac{3}{\pi^2} \log^2(x) + O(\log x)$$

Proof. We begin by considering $A(x) = \sum 2^{\omega(n)}$ and see

$$\begin{aligned} A(x) &= \sum_{n \leq x} 2^{\omega(n)} = \sum_{n \leq x} \sum_{d|n} \mu^2(d) \\ &= \sum_{d \leq x} \mu^2(d) \sum_{\substack{n \leq x \\ d|n}} 1 \\ &= \sum_{d \leq x} \mu^2(d) \left(\frac{x}{d} + O(1) \right) \\ &= x \left(\sum_{d \leq x} \mu^2(d)/d \right) + O\left(\sum_{d \leq x} \mu^2(d) \right) \end{aligned}$$

and we know that $\sum_{d \leq x} \mu^2(d)/d = 6/\pi^2 \log x + O(1)$ so we have

$$\begin{aligned} A(x) &= x \left(\frac{6}{\pi^2} \log x + O(1) \right) + O(1) \\ &= \frac{6}{\pi^2} x \log x + O(x) \end{aligned}$$

but we want $2^\omega(n)/n$, so we need to use partial summation. Letting $A(x)$ be as before, $f(x) = 1/x$ and $S(x) = \sum 2^{\omega(n)}/n$ we get

$$\begin{aligned} S(x) &= \frac{A(x)}{x} + \int_1^x \frac{S(t)}{t^2} dt \\ &= \frac{6}{\pi^2} \log x + O(1) + \int_1^x \left(\frac{6 \log t}{\pi^2 t} + O\left(\frac{1}{t}\right) \right) dt \\ &= \frac{6}{\pi^2} \log x + O(1) + \frac{3}{\pi^2} \log^2(x) + O(\log x) \\ &= \frac{3}{\pi^2} \log^2(x) + O(\log x) \end{aligned}$$

□

Lemma 5. *We have that*

$$\frac{1}{\varphi(n)} \ll \frac{\log \log(n)}{n}$$

Proof. Begin by considering $\log \frac{n}{\varphi(n)}$. By definition of $\varphi(n)$, this becomes

$$\begin{aligned} \log \left(\prod_{p|n} \left(1 - \frac{1}{p} \right)^{-1} \right) &= - \sum_{p|n} \log \left(1 - \frac{1}{p} \right) \\ &= \sum_{p|n} \left(\frac{1}{p} + O\left(\frac{1}{p^2}\right) \right) \\ &\ll \sum_{p \leq n} \frac{1}{p} \ll \log \log(n) \end{aligned}$$

where in line 2 we used Taylor, and in the last line we used (3). □

Lemma 6. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} \frac{1}{\varphi(n)} = A \log x + O\left(\frac{\log x \log \log x}{x}\right)$$

Proof. The strategy is to use a Dirichlet convolution. First we find the convolution for $\frac{n}{\varphi(n)}$. We let $f(n) = n/\varphi(n) = \sum_{d|n} g(d)$ and compute

$$f(p^k) = \frac{p^k}{p^{k-1}(p-1)} = \frac{p}{p-1} = 1 + \frac{1}{p-1}$$

Now we solve for g . We have

$$\begin{aligned} f(1) &= g(1) \implies g(1) = 1 \\ f(p) &= g(1) + g(p) \implies g(p) = \frac{1}{p-1} \end{aligned}$$

and for any higher power, $f(p^k)$ is constant, so $g(p^k)$ will be zero. Now we note that this means g is only nonzero on squarefree numbers. Thus we can

observe

$$g(n) = \frac{\mu^2(n)}{\varphi(n)}$$

now upon denoting $S(x) = \sum_{n \leq x} \frac{1}{\varphi(n)}$ we have

$$\begin{aligned} S(x) &= \sum_{n \leq x} \frac{1}{n} \sum_{d|n} \left(\frac{\mu^2(n)}{\varphi(n)} \right) \\ &= \sum_{d \leq x} \frac{\mu^2(d)}{d\varphi(d)} \sum_{m \leq x/d} \frac{1}{m} \\ &= \sum_{d \leq x} \frac{\mu^2(d)}{d\varphi(d)} \left(\log \frac{x}{d} + O(1) \right) \end{aligned}$$

now we have $S(x) = \log x M_1(x) + M_2(x) + M_3(x)$ where

$$\begin{aligned} M_1(x) &= \left(\sum_{d \leq x} \frac{g(d)}{d} \right) \\ M_2(x) &= - \sum_{d \leq x} \frac{g(d) \log d}{d} \\ M_3 &= O \left(\sum_{d \leq x} \frac{g(d)}{d} \right) \end{aligned}$$

First we evaluate $M_1(x)$:

$$\begin{aligned} M_1(x) &= \sum_{d=1}^{\infty} \frac{g(d)}{d} - \sum_{d > x} \frac{g(d)}{d} \\ &= \prod_p \left(1 + \frac{1}{p\varphi(p)} \right) - \sum_{d > x} \frac{g(d)}{d} \\ &= \prod_p \frac{(1 - p^{-2})^{-1} (1 - p^{-3})^{-1}}{(1 - p^{-6})^{-1}} - \sum_{d > x} \frac{g(d)}{d} \\ &= \frac{\zeta(2)\zeta(3)}{\zeta(6)} - \sum_{d > x} \frac{g(d)}{d} \\ &= \frac{\zeta(2)\zeta(3)}{\zeta(6)} + O \left(\frac{\log \log x}{x} \right) \end{aligned}$$

where in the last line we used 5. Now we evaluate $M_2(x)$:

$$\begin{aligned} M_2(x) &= - \sum_{d=1}^{\infty} \frac{g(d) \log d}{d} - \sum_{d \geq x} \frac{g(d) \log d}{d} \\ &= - \sum_{d=1}^{\infty} \frac{g(d) \log d}{d} + O\left(\frac{\log x \log \log x}{x}\right) \end{aligned}$$

via the same logic as $M_1(x)$. Lastly we evaluate $M_3(x)$:

$$M_3(x) \ll \frac{\log \log x}{x}$$

thus we get

$$\begin{aligned} S(x) &= \log x \left(A + O\left(\frac{\log \log x}{x}\right) \right) + O\left(\frac{\log \log x}{x}\right) \\ &= A \log x + O\left(\frac{\log x \log \log x}{x}\right) \end{aligned}$$

where $A = \frac{\zeta(2)\zeta(3)}{\zeta(6)}$ and $C = \sum_{d=1}^{\infty} \frac{g(d) \log d}{d}$. □

Lemma 7. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} \sigma(n) = \frac{\zeta(2)}{2} x^2 + O(x \log x) = \frac{\pi^2}{12} x^2 + O(x \log x).$$

Proof. We begin by considering

$$\begin{aligned} \sum_{n \leq x} \frac{\sigma(n)}{n} &= \sum_{n \leq x} \frac{1}{n} \sum_{d|n} d = \sum_{n \leq x} \sum_{d|n} \frac{d}{n} = \sum_{n \leq x} \sum_{m=(n/d)|n} \frac{1}{m} \\ &= \sum_{m \leq x} \frac{1}{m} \sum_{\substack{n \leq x \\ n \equiv 0 \pmod{m}}} 1 = \sum_{m \leq x} \frac{1}{m} \left(\frac{x}{m} + O(1) \right) \\ &= x \sum_{m \leq x} \frac{1}{m^2} + O(\log x) = x\zeta(2) + O(\log x), \end{aligned}$$

where in the last line we used

$$\sum_{m \leq x} \frac{1}{m^2} = \sum_{m=1}^{\infty} \frac{1}{m^2} - \sum_{m > x} \frac{1}{m^2} = \zeta(2) + O\left(\int_x^{\infty} \frac{dt}{t^2}\right) = \zeta(2) + O\left(\frac{1}{x}\right).$$

Then by partial summation we get

$$\begin{aligned} \sum_{n \leq x} \sigma(n) &= x \sum_{n \leq x} \frac{\sigma(n)}{n} - \int_1^x \left(\sum_{n \leq t} \frac{\sigma(n)}{n} \right) dt \\ &= x^2 \zeta(2) - \zeta(2) \int_1^x t dt + O(x \log x) = \frac{\zeta(2)}{2} x^2 + O(x \log x). \end{aligned}$$

□

Remark 1. One can strengthen 7 to

$$\sum_{n \leq x} \sigma(n) = \frac{\zeta(2)}{2} x^2 - \frac{1}{2} x \log x + O(x)$$

Proof. The general idea for the proof is to use the hyperbola method as with the divisor function $d(n)$, and reduce the length to $\sqrt{x}$. Begin by seeing:

$$\begin{aligned} \sum_{n \leq x} \sigma(n) &= \sum_{n \leq x} \sum_{d|n} d \\ &= \sum_{n \leq x} \sum_{ab|n} a = \sum_{ab \leq x} a \\ &= \sum_{a \leq \sqrt{x}} \sum_{b \leq x/a} a + \sum_{b \leq \sqrt{x}} \sum_{a \leq x/b} a - \sum_{a \leq \sqrt{x}} \sum_{b \leq \sqrt{x}} a \end{aligned}$$

just as in the hyperbola method for $d(n)$ we split into 3 sums, where the last one is to correct for over-counting in the first two. Upon denoting the three sums $S_1(x), S_2(x), S_3(x)$ respectively, we see

$$\begin{aligned} S_1(x) &= \sum_{a \leq \sqrt{x}} \sum_{b \leq x/a} a \\ &= \sum_{a \leq x} a \sum_{b \leq x/a} 1 \\ &= \sum_{a \leq \sqrt{x}} a \left(\frac{[x/a]([x/a] + 1)}{2} \right) \\ &= \frac{1}{2} \sum_{a \leq \sqrt{x}} a \left(\left(\frac{x}{a} \right)^2 + O\left(\frac{x}{a} \right) \right) \\ &= \frac{x^2}{2} \sum_{a \leq \sqrt{x}} \frac{1}{a} + O\left(x \sum_{a \leq \sqrt{x}} 1 \right) \end{aligned}$$

□

Lemma 8. Let

$$f(n) = \sum_{d|n} g(d)$$

where $g(d)$ is completely multiplicative. We then have the identity:

$$f(mp) = (1 + g(p))f(m) - f(m/p).$$

Proof. We consider cases. If $(m, p) = 1$ then we have:

$$f(mp) = f(m)f(p) = f(m) \sum_{d|p} g(d) = f(m)(g(1) + g(p)) = (1 + g(p))f(m).$$

Defining $f(m/p) = 0$ when $m/p \notin \mathbb{Z}$ gives the desired formula. Now if $(m, p) > 1$, then we have:

$$\begin{aligned} f(mp) &= \sum_{d|mp} g(d) = \sum_{d|m} g(d) + \sum_{g|m} g(dp) - \sum_{d|\frac{m}{p}} g(d) \\ &= (1 + g(p)) \sum_{d|m} g(d) - \sum_{d|\frac{m}{p}} g(d) = (1 + g(p))f(m) - f(m/p). \end{aligned}$$

□

However, as $\sigma(n)$ is multiplicative, but not completely multiplicative the Lemma above is not applicable to σ . Instead, we deduce the following particular cases.

Lemma 9. *For any primes p, q and any positive integer m we have*

$$\sigma(mp) = (p+1)\sigma(m) - p\sigma\left(\frac{m}{p}\right)$$

and

$$\sigma(mpq) = (p+1)(q+1)\sigma(m) - (p+1)q\sigma\left(\frac{m}{q}\right) - p(q+1)\sigma\left(\frac{m}{p}\right) + pq\sigma\left(\frac{m}{pq}\right).$$

Proof. For $(m, p) = 1$ we have

$$\sigma(mp) = \sigma(m)\sigma(p) = (p+1)\sigma(m),$$

which satisfies the required formula as $m/p \notin \mathbb{N}$. For $(m, p) > 1$ we classify the divisors of mp into three groups: the divisors of m , the divisors of m multiplied by p and the intersection of the first two. Then by inclusion-exclusion we have

$$\sigma(mp) = \sum_{d|mp} d = \sum_{d|m} d + \sum_{d|m} pd - \sum_{d|m/p} pd = (p+1)\sigma(m) - p\sigma(m/p).$$

On iterating that twice we get

$$\begin{aligned} \sigma(mpq) &= (q+1)\sigma(mp) - q\sigma\left(\frac{mp}{q}\right) \\ &= (q+1)\left((p+1)\sigma(m) - p\sigma\left(\frac{m}{p}\right)\right) - q\left((p+1)\sigma\left(\frac{m}{q}\right) - p\sigma\left(\frac{m}{pq}\right)\right) \\ &= (p+1)(q+1)\sigma(m) - (p+1)q\sigma\left(\frac{m}{q}\right) - p(q+1)\sigma\left(\frac{m}{p}\right) + pq\sigma\left(\frac{m}{pq}\right). \end{aligned}$$

□

Lemma 10. *For any prime $p, q \equiv 1 \pmod{4}$ and any positive integer m we have*

$$r_2(mp) = 2r_2(m) - r_2(m/p)$$

and

$$r_2(mpq) = 4r_2(m) - 2r_2\left(\frac{m}{p}\right) - 2r_2\left(\frac{m}{q}\right) + r_2\left(\frac{m}{pq}\right).$$

Proof. The proof of the previous lemma follows from Lemma 8, along with iteration by considering $r_2(mpq) = r_2((mp)q) = 2r_2(mp) - r_2((m/q)p)$. $\square$

Lemma 11. *As $x \rightarrow \infty$ we have*

$$\sum_{p \leq x} \frac{1}{\sqrt{p}} \ll \sqrt{x},$$

$$\sum_{pq \leq x} \frac{1}{\sqrt{p}\sqrt{q}} \ll \sqrt{x} \log \log x$$

Proof. We begin with the first claim. Using prime number theorem in the form

$$\pi(x) = \sum_{p \leq x} 1 = \frac{x}{\log x} + O\left(\frac{x}{\log^2 x}\right)$$

and partial summation we get

$$\begin{aligned} \sum_{p \leq x} \frac{1}{\sqrt{p}} &= \frac{\pi(x)}{\sqrt{x}} + \frac{1}{2} \int_2^x \pi(t) \frac{dt}{t^{3/2}} \\ &= \frac{\sqrt{x}}{\log x} + O\left(\frac{\sqrt{x}}{\log^2 x}\right) + \int_2^x \left(\frac{t}{\log t} + O\left(\frac{t}{\log^2 t}\right)\right) \frac{dt}{t^{3/2}} \\ &= \frac{\sqrt{x}}{\log x} + \frac{1}{2} \int_2^x \frac{dt}{\sqrt{t} \log t} + O\left(\frac{\sqrt{x}}{\log^2 x} + \int_2^x \frac{dt}{\sqrt{t} \log^2 t}\right). \end{aligned}$$

We further notice that

$$\int_2^x \frac{dt}{\sqrt{t} \log^2 t} \ll \sqrt{x} \int_2^x \frac{dt}{t \log^2 t} = \sqrt{x} \int_2^x \frac{d \log t}{\log^2 t} = \sqrt{x} \left(-\frac{1}{\log t}\right)_2^x \ll \sqrt{x}.$$

Integrating by parts we get

$$\frac{1}{2} \int_2^x \frac{dt}{\sqrt{t} \log t} = \int_2^x \frac{d(\sqrt{t})}{\log t} = \frac{\sqrt{t}}{\log t} \Big|_2^x + \int_2^x \frac{\sqrt{t}}{t \log^2 t} dt \ll \sqrt{x}.$$

Therefore

$$\sum_{p \leq x} \frac{1}{\sqrt{p}} \ll \sqrt{x}.$$

Now using the first part, we finish the second claim. Observe

$$\begin{aligned} \sum_{p, q \leq x} \frac{1}{\sqrt{p}\sqrt{q}} &= \sum_{p \leq x} \frac{1}{\sqrt{p}} \sum_{q \leq x/p} \frac{1}{\sqrt{q}} \ll \sum_{p \leq x} \frac{1}{\sqrt{p}} \sqrt{\frac{x}{p}} \ll \sqrt{x} \sum_{p \leq x} \frac{1}{p} \\ &\ll \sqrt{x} \log \log x \end{aligned}$$

by Mertens' theorem (3). $\square$

Lemma 12. *Mean value of $r_1(n)$*

As $x \rightarrow \infty$ we have

$$\sum_{n \leq x} r_1(n) = \frac{\pi x}{2 \log x} + O(\sqrt{x})$$

Proof. We begin by denoting the sum in question $S(x)$ simply expanding.

$$\begin{aligned} S(x) &= \sum_{n \leq x} r_1(n) = \sum_{a^2 + p^2 \leq x} 1 \\ &= \sum_{a \leq \sqrt{x}} \sum_{p \leq \sqrt{x-a^2}} 1 \end{aligned}$$

and then apply PNT to obtain

$$S(x) = \sum_{a \leq \sqrt{x}} \left(\int_2^{\sqrt{x-a^2}} \frac{dt}{\log t} + O\left(\sqrt{x-a^2} e^{-C\sqrt{\log \sqrt{x-a^2}}}\right) \right)$$

□

2. SUMS OF TWO SQUARES

2.1. Mean value of $r_2(n)\omega(n)$.

Lemma 13. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} r_2(n)\omega(n) = \frac{\pi}{4} x \log \log x + O(x).$$

Proof. Let $S(x)$ denote the sum in question and proceed as usual.

$$\begin{aligned} S(x) &= \sum_{n \leq x} r_2(n)\omega(n) = \sum_{n \leq x} r_2(n) \sum_{p|n} 1 \\ &= \sum_{p \leq x} \sum_{m \leq x/p} r_2(mp) \end{aligned}$$

Now we must be careful since we can only apply (10) when $p \equiv 1(4)$. However, we can show that the terms when $p \equiv 3(4)$ do not contribute to the estimation. Consider this case

$$\begin{aligned} \sum_{\substack{p \leq x \\ p \not\equiv 1(4)}} \sum_{m \leq x/p} r_2(mp) &= \sum_{m \leq x/2} (r_2(m) - r_2(m/2)) - \sum_{\substack{p \leq x \\ p \equiv 3(4)}} \sum_{m \leq x/p} r_2(m/p) \\ &= \frac{\pi}{4} \frac{x}{2} + O(\sqrt{x}) - \frac{\pi}{4} \frac{x}{4} + O(\sqrt{x}) - \sum_{\substack{p \leq x \\ p \equiv 3(4)}} \sum_{m \leq x/p^2} r_2(m/p) \\ &\ll x + \sum_{\substack{p \leq x \\ p \equiv 3(4)}} \frac{x}{p^2} \ll x \end{aligned}$$

since we have a $O(x)$ in our final estimation, we can safely ignore any primes not congruent to 1 mod 4. Thus, using (10) we get

$$\begin{aligned}
S(x) &= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \sum_{m \leq x/p} r_2(mp) \\
&= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \sum_{m \leq x/p} 2r_2(m) - r_2(m/p) \\
&= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \sum_{m \leq x/p} 2r_2(m) - \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \sum_{u=m/p \leq x/p^2} r_2(u) \\
&= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} 2 \left(\frac{\pi}{4} \frac{x}{p} + O(\sqrt{x/p}) \right) - \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \left(\frac{\pi}{4} \frac{x}{p^2} + O\left(\frac{\sqrt{x}}{p}\right) \right),
\end{aligned}$$

where in the last line we used (1). For the last sum we have

$$\sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p^2} \ll \sum_{n=1}^{\infty} \frac{1}{n^2} \ll 1, \quad \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p} \ll \log \log x$$

and hence using (4) we get

$$S(x) = \frac{\pi x}{2} \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p} + O\left(\sqrt{x} \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{\sqrt{p}}\right) + O(x).$$

We apply Lemma 11 and conclude

$$S(x) = \frac{\pi x}{2} \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p} + O(x) = \frac{\pi x}{4} \log \log x + O(x),$$

where in the last equality we used Lemma 4. □

2.2. Mean value of $r_2(n)\omega^2(n)$.

Lemma 14. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} r_2(n)\omega^2(n) = \frac{\pi}{4} x (\log \log x)^2 + O(x \log \log x).$$

Proof. Proceed as usual. Let $S(x)$ denote the sum in question. We have:

$$\begin{aligned}
S(x) &= \sum_{n \leq x} r_2(n)\omega^2(n) = \sum_{n \leq x} r_2(n) \sum_{p|n} 1 \sum_{q|n} 1 \\
&= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{\substack{n \leq x \\ n \equiv 0(p) \\ n \equiv 0(q)}} r_2(n) + \sum_{p \leq x} \sum_{\substack{n \leq x \\ n \equiv 0(p)}} r_2(n) \\
&= S_1(x) + x \log \log x + O(x),
\end{aligned}$$

where we used that

$$\sum_{p \leq x} \sum_{\substack{n \leq x \\ n \equiv 0(p)}} r_2(n) = \sum_{n \leq x} r_2(n) \omega(n) = \frac{\pi}{4} x \log \log x + O(x)$$

by Lemma 13. Therefore it remains to evaluate $S_1(x)$. We have

$$\begin{aligned} S_1(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{\substack{n \leq x \\ n \equiv 0(p) \\ n \equiv 0(q)}} r_2(n) = \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2(mpq) \\ &= 4M_1(x) - 2M_2(x) - 2M_3(x) + M_4(x), \end{aligned}$$

where we used (10) to expand $r_2(mpq)$ and denoted

$$\begin{aligned} M_1(x) &= \sum_{\substack{pq \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2(m), \\ M_2(x) &= \sum_{\substack{pq \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2\left(\frac{m}{p}\right), \\ M_3(x) &= \sum_{\substack{pq \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2\left(\frac{m}{q}\right), \\ M_4(x) &= \sum_{\substack{pq \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2\left(\frac{m}{pq}\right). \end{aligned}$$

Note that the condition $pq \leq x$ is stronger than $p, q \leq x$. This condition arises as for $n \leq x$ with $n \equiv 0 \pmod{p}$, $n \equiv 0 \pmod{q}$, $p \neq q$ we have $n = mpq \leq x$, which implies $pq \leq x$. Another way to see is to apply Chinese remainder theorem directly

$$\sum_{n \leq x} r_2(n) \sum_{p|n} 1 \sum_{q|n} 1 = \sum_{\substack{\text{lcm}(p,q) \leq x \\ n \equiv 0 \pmod{\text{lcm}(p,q)}}} \sum_{n \leq x} r_2(n)$$

and notice that $\text{lcm}(p, q) = pq$ for $p \neq q$ and $\text{lcm}(p, q) = p$ for $p = q$. Focusing on $M_1(x)$ we have

$$M_1(x) = \sum_{\substack{pq \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2(m) = \sum_{pq \leq x} \sum_{m \leq x/(pq)} r_2(m) - \sum_{p \leq x} \sum_{m \leq x/p^2} r_2(m).$$

Notice that the second sum is bounded by (1) as

$$\sum_{p \leq x} \sum_{m \leq x/p^2} r_2(m) \ll \sum_{p \leq x} \frac{x}{p^2} \ll x.$$

Thus it is enough to focus on the first sum. Using (1) we write

$$\begin{aligned}
 \sum_{pq \leq x} \sum_{m \leq x/(pq)} r_2(m) &= \frac{\pi x}{4} \sum_{pq \leq x} \left(\frac{1}{pq} + O\left(\frac{\sqrt{x}}{\sqrt{pq}}\right) \right) \\
 &= \frac{\pi x}{4} (\log \log x)^2 + O\left(\sqrt{x} \sum_{pq \leq x} \frac{1}{\sqrt{pq}}\right) \\
 &= \frac{\pi x}{4} (\log \log x)^2 + O(x \log \log x),
 \end{aligned}$$

where in the second line we used Lemma 1 and in the last line we used (11). Now we turn to evaluate $M_2(x)$ (note this is also evaluating $M_3(x)$ via symmetry).

$$\begin{aligned}
 M_2(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2\left(\frac{m}{p}\right) = \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{u=m/p \leq x/(p^2q)} r_2(u) \\
 &= \sum_{\substack{p, q \leq x \\ p \neq q}} \left(\frac{\pi x}{4p^2q} + O(\sqrt{x/(p^2q)}) \right) \\
 &\ll x \log \log x
 \end{aligned}$$

so it goes to the error term. Similarly for $M_4(x)$ we have

$$\begin{aligned}
 M_4(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} r_2\left(\frac{m}{pq}\right) = \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{u=m/(pq) \leq x/(pq)^2} r_2(u) \\
 &= \sum_{\substack{p, q \leq x \\ p \neq q}} \frac{\pi x}{4(pq)^2} + O(\sqrt{x/(pq)^2}) \\
 &\ll x \log \log x
 \end{aligned}$$

hence it also goes to the error. Thus for the final estimate we have:

$$S(x) = \frac{\pi x}{4} (\log \log x)^2 + O(x \log \log x).$$

□

3. SUMS OF FOUR SQUARES

3.1. Mean value of $r_4(n)\omega(n)$.

Lemma 15. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} r_4(n)\omega(n) = \frac{\pi^2}{12} \frac{15}{16} x^2 \log \log x + O(x^2).$$

Proof. By definition $r_4(n) = \sigma(n) - \sigma(n/4)$ so just plugging this into the equation and denoting the sum in question by $S(x)$ we get

$$\begin{aligned} S(x) &= \sum_{n \leq x} (\sigma(n) - \sigma(n/4)) \omega(n) \\ &= \sum_{n \leq x} \sigma(n) \omega(n) - \sum_{n \leq x} \sigma(n/4) \omega(n) = S_1(x) - S_2(x). \end{aligned}$$

On changing variables in $S_2(x)$ we get

$$\begin{aligned} S_2(x) &= \sum_{n \leq x} \sigma(n/4) \omega(n) = \sum_{m \leq x/4} \sigma(m) \omega(4m) = \sum_{m \leq x/4} \sigma(m) (\omega(m) + 1_{2 \nmid m}) \\ &= S_1(x/4) + \sum_{\substack{m \leq x/4 \\ 2 \nmid m}} \sigma(m), \end{aligned}$$

where we used that

$$\begin{aligned} (5) \quad \omega(4m) &= \sum_{p|4m} 1 = \omega(m) + \omega(4) - \omega(\gcd(m, 4)) = \omega(m) + 1 - 1_{2|m} \\ &= \omega(m) + 1_{2 \nmid m}. \end{aligned}$$

Thus,

$$S(x) = S_1(x) - S_1(x/4) - \sum_{\substack{n \leq x \\ 2 \nmid n}} \sigma(n) = S_1(x) - S_1(x/4) + O(x^2)$$

via using Lemma 7. Hence it is enough to compute $S_1(x)$. We have

$$\begin{aligned} S_1(x) &= \sum_{n \leq x} \sigma(n) \sum_{p|n} 1 = \sum_{p \leq x} \sum_{m \leq x/p} \sigma(mp) \\ &= \sum_{p \leq x} \sum_{m \leq x/p} \left((p+1)\sigma(m) - p\sigma(m/p) \right) \\ &= \sum_{p \leq x} (p+1) \sum_{m \leq x/p} \sigma(m) - \sum_{p \leq x} p \sum_{u=m/p \leq x/p^2} \sigma(u). \end{aligned}$$

Using Lemma 7 for the first summand we get

$$\begin{aligned} \sum_{p \leq x} (p+1) \sum_{m \leq x/p} \sigma(m) &= \sum_{p \leq x} (p+1) \left(\frac{\pi^2}{12} \frac{x^2}{p^2} + O\left(\frac{x}{p} \log \frac{x}{p}\right) \right) \\ &= \frac{\pi^2}{12} x^2 \sum_{p \leq x} \frac{p+1}{p^2} + O(x \log x \log \log x), \end{aligned}$$

where in the error term we used (3) and (2) to get

$$\begin{aligned} \sum_{p \leq x} \frac{\log(x/p)}{p} &= \log x \sum_{p \leq x} \frac{1}{p} - \sum_{p \leq x} \frac{\log p}{p} \\ &= \log x \log \log x + (A - 1) \log x + O\left(\frac{1}{\log x}\right). \end{aligned}$$

Further, in the first sum we apply Mertens' theorem (3) to get

$$\begin{aligned} \sum_{p \leq x} \frac{p+1}{p^2} &= \sum_{p \leq x} \frac{1}{p} + \sum_{p \leq x} \frac{1}{p^2} = \sum_{p \leq x} \frac{1}{p} + \sum_{p \geq 2} \frac{1}{p^2} - \sum_{p > x} \frac{1}{p^2} \\ &= \log \log x + A + C_2 + O\left(\frac{1}{\log x}\right), \end{aligned}$$

where

$$C_2 = \sum_{p \geq 2} \frac{1}{p^2} \ll \sum_{n \geq 2} \frac{1}{n^2} \ll 1$$

and the tail is bounded as

$$\sum_{p > x} \frac{1}{p^2} \ll \sum_{n > x} \frac{1}{n^2} \ll \int_x^\infty \frac{dt}{t^2} \ll \frac{1}{x}.$$

Working in a similar manner with the second summand in $S_1(x)$ we find that

$$\begin{aligned} \sum_{p \leq x} p \sum_{u \leq x/p^2} \sigma(u) &= \sum_{p \leq x} p \left(\frac{x^2}{p^4} + O\left(\frac{x}{p^2} \log \frac{x}{p^2}\right) \right) \\ &= x^2 \sum_{p \leq x} \frac{1}{p^3} + O\left(x \log x \log \log x\right) \end{aligned}$$

as

$$\sum_{p \leq x} \frac{\log(x/p^2)}{p} \ll \log x \sum_{p \leq x} \frac{1}{p} \ll \log x \log \log x.$$

Further, defining

$$\sum_{p \leq x} \frac{1}{p^3} = \sum_{p \geq 2} \frac{1}{p^3} + O\left(\int_x^\infty \frac{dt}{t^3}\right) = C_3 + O\left(\frac{1}{x^2}\right)$$

we get

$$\sum_{p \leq x} p \sum_{u \leq x/p^2} \sigma(u) = C_3 x^2 + O(x \log x \log \log x).$$

Therefore, we conclude

$$\begin{aligned} S_1(x) &= \frac{\pi^2}{12} x^2 \log \log x + (A + C_2 + C_3) x^2 + O\left(\frac{x^2}{\log x}\right) \\ &= \frac{\pi^2}{12} x^2 \log \log x + O(x^2). \end{aligned}$$

Therefore

$$\begin{aligned} S_1(x) - S_1(x/4) &= \frac{\pi^2}{12} x^2 \left(\log \log x - \frac{1}{16} \log \log \frac{x}{4} \right) + O(x^2) \\ &= \frac{\pi^2}{12} \frac{15}{16} x^2 \log \log x + O(x^2) \end{aligned}$$

via

$$\begin{aligned} \log \log \frac{x}{4} &= \log \left(\log x \left(1 - \frac{\log 4}{\log x} \right) \right) = \log \log x + \log \left(1 - \frac{\log 4}{\log x} \right) \\ &= \log \log x + O\left(\frac{1}{\log x} \right). \end{aligned}$$

□

3.2. Mean value of $r_4(n)\omega^2(n)$.

Lemma 16. *As $x \rightarrow \infty$ we have*

$$\sum_{n \leq x} r_4(n) \omega(n)^2 = \frac{\pi^2}{12} \frac{15}{16} x^2 (\log \log x)^2 + O(x^2 \log \log x),$$

Proof. By definition, $r_4 = \sigma(n) - \sigma(n/4)$ so plug this into the equation and denote the desired sum as $S(x)$ to get

$$\begin{aligned} S(x) &= \sum_{n \leq x} (\sigma(n) - \sigma(n/4)) \omega^2(n) \\ &= \sum_{n \leq x} \sigma(n) \omega^2(n) - \sum_{n \leq x} \sigma(n/4) \omega^2(n) = S_1(x) - S_2(x). \end{aligned}$$

Changing variables in $S_2(x)$ gives

$$\begin{aligned} S_2(x) &= \sum_{n \leq x} \sigma(n/4) \omega^2(n) = \sum_{m \leq x/4} \sigma(m) \omega^2(4m) \\ &= \sum_{m \leq x/4} \sigma(m) \omega^2(m) + 2 \sum_{\substack{m \leq x/4 \\ 2 \nmid m}} \sigma(m) \omega(m) + \sum_{\substack{m \leq x/4 \\ 2 \nmid m}} \sigma(m) \\ &= S_1(x/4) + O(x^2 \log \log x), \end{aligned}$$

where we used (5) to show that

$$\omega^2(4m) = (\omega(m) + 1_{2 \nmid m})^2 = \omega^2(m) + 2\omega(m)1_{2 \nmid m} + 1_{2 \nmid m}.$$

We also use Lemma 15 to bound the $\sigma(m)\omega(m)$ term by $x^2 \log \log x$ and Lemma 7 to bound the $\sigma(m)$ sum by x^2 . Thus,

$$S(x) = S_1(x) - S_1(x/4) + O(x^2 \log \log x).$$

Hence it is enough to compute $S_1(x)$. We have:

$$\begin{aligned}
S_1(x) &= \sum_{n \leq x} \sigma(n) \sum_{p|n} 1 \sum_{q|n} 1 = \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{\substack{n \leq x \\ n \equiv 0(p) \\ n \equiv 0(q)}} \sigma(n) + \sum_{p \leq x} \sum_{\substack{n \leq x \\ n \equiv 0(p)}} \sigma(n) \\
&= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{\substack{n \leq x \\ n \equiv 0(p) \\ n \equiv 0(q)}} \sigma(n) + \sum_{n \leq x} \sigma(n) \omega(n) \\
&= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{\substack{n \leq x \\ n \equiv 0(p) \\ n \equiv 0(q)}} \sigma(n) + O(x^2 \log \log x)
\end{aligned}$$

via Lemma 15 for the error term. Focus on the first term and change order of summation to get

$$\sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{\substack{n \leq x \\ n \equiv 0(p) \\ n \equiv 0(q)}} \sigma(n) = \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} \sigma(mpq) = M_1(x) - M_2(x) - M_3(x) + M_4(x).$$

where we used Lemma 9 to expand mpq and denoted

$$\begin{aligned}
M_1(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} (p+1)(q+1)\sigma(m), \\
M_2(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} (p+1)q\sigma\left(\frac{m}{q}\right), \\
M_3(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} p(q+1)\sigma\left(\frac{m}{p}\right), \\
M_4(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} pq\sigma\left(\frac{m}{pq}\right).
\end{aligned}$$

Now we must be careful about error terms. We are going to show that $M_1(x)$ contains the main term, while $M_2(x)$, $M_3(x)$, $M_4(x)$ all go to the error. Using Lemma 7 we first write $M_1(x)$ as

$$\begin{aligned}
M_1(x) &= \sum_{p, q \leq x} \sum_{m \leq x/(pq)} (p+1)(q+1)\sigma(m) - \sum_{p \leq x} \sum_{m \leq x/p^2} (p+1)^2\sigma(m) \\
&= \sum_{p, q \leq x} (p+1)(q+1) \left(\frac{\pi^2}{12} \frac{x^2}{p^2 q^2} + O\left(\frac{x}{pq} \log \frac{x}{pq}\right) \right) + O\left(\sum_{p \leq x} (p+1)^2 \frac{x^2}{p^4}\right) \\
&= \frac{\pi^2 x^2}{12} \left(\sum_{p \leq x} \frac{p+1}{p^2} \right)^2 + O\left(x \log x \left(\sum_{p \leq x} \frac{1}{p} \right)^2 + x^2\right),
\end{aligned}$$

where we used that $\log p, \log q \ll \log x$ to bound the error term. Further, by Mertens' theorem (3)

$$\sum_{p \leq x} \frac{p+1}{p^2} = \sum_{p \leq x} \frac{1}{p} + O(1) = \log \log x + O(1)$$

and hence

$$M_1(x) = \frac{\pi^2}{12} x^2 (\log \log x)^2 + O(x^2 \log \log x).$$

The terms $M_2(x)$ and $M_3(x)$ can be bounded in the same way via changing the roles of p and q , hence we consider $M_2(x)$ only.

$$\begin{aligned} M_2(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} (p+1)q \sigma\left(\frac{m}{q}\right) \ll \sum_{p, q \leq x} (p+1)q \sum_{u=m/q \leq x/(pq^2)} \sigma(u) \\ &\ll \sum_{p, q \leq x} (p+1)q \left(\frac{x}{pq^2}\right)^2 \ll x^2 \sum_{p \leq x} \frac{1}{p} \sum_{q \leq x} \frac{1}{q^3} \ll x^2 \log \log x. \end{aligned}$$

Finally, $M_4(x)$ can be bounded similarly as

$$\begin{aligned} M_4(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} pq \sigma\left(\frac{m}{pq}\right) \ll \sum_{p, q \leq x} pq \sum_{u=m/(pq) \leq x/(pq)^2} \sigma(u) \\ &\ll \sum_{p, q \leq x} pq \left(\frac{x}{(pq)^2}\right)^2 \ll x^2 \left(\sum_{p \leq x} \frac{1}{p^3}\right)^2 \ll x^2. \end{aligned}$$

On combining this we get

$$S_1(x) = \frac{\pi^2}{12} x^2 (\log \log x)^2 + O(x^2 \log \log x).$$

therefore

$$\begin{aligned} S(x) &= S_1(x) - S_1(x/4) + O(x^2 \log \log x) \\ &= \frac{\pi^2}{12} x^2 (\log \log x)^2 - \frac{\pi^2}{12} \left(\frac{x^2}{4^2}\right) \left(\log \log \left(\frac{x}{4}\right)\right)^2 + O(x^2 \log \log x) \\ &= \frac{\pi^2}{12} x^2 (\log \log x)^2 \left(1 - \frac{1}{16}\right) + O(x^2 \log \log x) \\ &= \frac{\pi^2}{12} \frac{15}{16} x^2 (\log \log x)^2 + O(x^2 \log \log x) \end{aligned}$$

□

Remark 2. One can strengthen Lemma 16 to

$$\sum_{n \leq x} r_4(n) \omega(n)^2 = \frac{\pi^2}{12} \frac{15}{16} x^2 (\log \log x)^2 + x^2 \log \log x + O(x^2).$$

Proof. We proceed as in Lemma 16 and notice that $M_4(x) \ll x^2$, hence we need to deal with $M_1(x)$, $M_2(x)$, $M_3(x)$ only. We need to get an asymptotic formula with two terms for $M_1(x)$ and with one term for $M_2(x)$, $M_3(x)$. As $M_2(x)$ and $M_3(x)$ are the same via changing the roles of p and q , we may consider $M_2(x)$ only.

$$\begin{aligned} M_2(x) &= \sum_{\substack{p, q \leq x \\ p \neq q}} \sum_{m \leq x/(pq)} (p+1)q\sigma\left(\frac{m}{q}\right) = \sum_{\substack{p, q \leq x \\ p \neq q}} (p+1)q \sum_{u \leq x/(pq^2)} \sigma(u) \\ &= \sum_{p, q \leq x} (p+1)q \sum_{u \leq x/(pq^2)} \sigma(u) - \sum_{p \leq x} (p+1)p \sum_{u \leq x/p^3} \sigma(u) \\ &= M'_2(x) + O(x^2) \end{aligned}$$

as we have

$$\sum_{p \leq x} (p+1)p \sum_{u \leq x/p^3} \sigma(u) \ll x^2 \sum_{p \leq x} \frac{1}{p^4} \ll x^2.$$

Further, by Lemma 7, we have

$$\begin{aligned} M'_2(x) &= \sum_{p, q \leq x} (p+1)q \left(\frac{\pi^2}{12} \frac{x^2}{p^2 q^4} + O\left(\frac{x}{pq^2} \log \frac{x}{pq^2} \right) \right) \\ &= \frac{\pi^2}{12} x^2 \left(\sum_{p \leq x} \frac{p+1}{p^2} \right) \left(\sum_{q \leq x} \frac{1}{q^3} \right) + O\left(x \log x \left(\sum_{p \leq x} 1 \right) \left(\sum_{q \leq x} \frac{1}{q^2} \right) \right) \\ &= \frac{\pi^2}{12} x^2 \left(\log \log x + O(1) \right) \sum_{q \leq x} \frac{1}{q^3} + O(x^2). \end{aligned}$$

On denoting

$$C_3 = \sum_{p \geq 2} \frac{1}{p^3}$$

we get

$$\sum_{q \leq x} \frac{1}{q^3} = C_3 + O\left(\int_x^\infty \frac{dt}{t^3} \right) = C_3 + O\left(\frac{1}{x^2} \right)$$

and hence

$$M_2(x) = \frac{\pi^2}{12} C_3 x^2 \log \log x + O(x^2).$$

□